Compare the tooth rows

A zipper clicks together.

Name _____ Date _____

Use the printed counts. The illustrations are not exact data.

1. One sample zipper has 38 elements along a pocket. Another has 46. Put >, <, or = in the circle. Which is the greater count?

38 ○ 46

Show your thinking.

2. A long bag zipper has 72 elements. A short pouch zipper has 27. Put >, <, or = in the circle. Which is the greater count?

72 ○ 27

Show your thinking.

3. A zipper design gives the left tape 54 elements and the right tape 54 elements. Put >, <, or = in the circle. What shared count do the matching rows have?

54 ○ 54

Show your thinking.

Compare the tooth rows

A zipper clicks together.

Name _____ Date _____

Use the printed counts. The illustrations are not exact data.

4. A design note lists 43 zipper elements. What is 43 rounded to the nearest ten?

43 → nearest ten: ___

Show your thinking.

5. Another zipper design lists 67 elements. What is 67 rounded to the nearest ten?

67 → nearest ten: ___

Show your thinking.

6. A long zipper design lists 95 elements. What is 95 rounded to the nearest ten?

95 → nearest ten: ___

Show your thinking.

A moving boundary

Below the slider, two tooth rows stood apart. Above it, they made one joined row. Little wondered what happened inside.

Bump meets pocket

Each zipper element had a bump and a pocket. Left and right elements alternated, nesting across the middle to hold together.

A hidden Y path

Inside the slider, two channels curved around a center wedge and met. Angled walls guided both tooth rows inward.

One after another

The channel lined up left and right elements. Bump met pocket, then the next pair followed. A flexible zipper chain formed behind the slider.

The wedge separates

Moving back sent the joined row toward the center wedge. It guided one side left and the other right, opening the zipper.

Click at the stop

The last elements nested. The slider clicked against the top stop, which kept it from running off the closed row.

Make a little book

1 Fold it into 8 boxes. Open it flat.

2 Fold short end to short end. Cut the thick line. Open it.

3 Fold the long way, pictures out. Push the ends in.

4 Fold into a book. Page 1 goes on the front!

WONDER DAILY

A zipper clicks together.

Give the mechanism your words.

A zipper clicks together. · Curiosity day

Name _____ Date _____

Choose a scene. Look, say an idea, then draw or write.



Where open becomes closed



Bumps fit pockets



A Y inside the slider

Draw the slider's Y-shaped path. Add arrows to show two rows becoming one.

element · slider · channel · interlock · wedge · because · however

At the boundary, … what is separate and what is joined?

The teeth hold because … how do the bumps and pockets fit?

Inside the slider, … what does the Y-shaped channel guide?

Read your words aloud. Did you connect a shape to its job with because, so, or however?

A zipper clicks together.

01 What changes right where the slider sits?



A close imagined view. The zipper is enlarged to help us see.

Look closer. Below the slider are two separate rows. Above it is one joined row.

A little explanation. The slider does not glue the sides. It guides matching zipper teeth into an interlocking pattern.

Try this. Trace each open row toward the slider, then follow the single closed row away from it.

02 Why are zipper teeth shaped with bumps and pockets?



Zipper elements are enlarged and simplified so their nesting shapes are clear.

Look closer. A bump on one side settles into a pocket on the other side.

A little explanation. Alternating teeth nest across the gap. Their shapes help the two rows hold together.

Try this. Point to one left tooth, then the right tooth that fits beside it.

03 What path is hiding inside the slider?



A cutaway explanation of the hidden channel, enlarged and simplified.

Look closer. Two wide channels curve toward one narrow opening, like the arms of a Y.

A little explanation. The slider's Y-shaped channel bends both rows inward in the right order.

Try this. Draw a Y in the air. Bring two fingertips down its arms and out together.

04 How can the same slider open the zipper again?



The same hidden path can join the teeth or guide them apart.

Look closer. Moving the other way sends the joined row toward the fork in the channel.

A little explanation. The center wedge guides the interlocked teeth apart and sends each row out a different side.

Try this. Trace the joined row toward the fork, then split your fingers along two paths.